# Skills Summary

## Parameters, Speeds and Polar Limits Without the Classic Errors

*Use this reference while completing your worksheet or when studying.*

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## Skill 1 — The parameter is not a coordinate

$\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$ (top over bottom, in that order), evaluated at a value of $t$ — never at an $x$- or $y$-value.

- To use a *point*, first solve $x(t) = x_0$ for $t$, then keep only the roots that also give $y(t) = y_0$.
- Horizontal tangent: $dy/dt = 0$ with $dx/dt \neq 0$. Vertical tangent: the other way round.
- $dy/dx$ describes the curve; it is unchanged if the same path is traced at a different rate.

**Example:** For $x(t) = t^2 + t$ and $y(t) = t^3 + 2t$, find $\dfrac{dy}{dx}$ at $t = 2$.
**Step 1.** A parameter value is given outright, so no point-solving is needed — go straight to the quotient, top over bottom.
**Step 2.** $\dfrac{dy/dt}{dx/dt} = \dfrac{3t^2 + 2}{2t + 1}$, and at $t = 2$ that is $\dfrac{14}{5}$.
$\mathbf{\frac{14}{5}}$

**Try it:** For $x(t) = t^2 - t$ and $y(t) = t^3 + t$, find $\dfrac{dy}{dx}$ at $t = 3$.
check: $\mathbf{\frac{28}{5}}$

## Skill 2 — Speed is a magnitude, not a sum

Velocity $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ (a vector). Speed $= |\mathbf{v}(t)| = \sqrt{x'(t)^2 + y'(t)^2}$ (a scalar, never negative).

- Distance travelled $= \displaystyle\int_a^b \sqrt{x'^2 + y'^2}\, dt$ — the integral of speed, not of the components.
- Displacement uses the components and can cancel; distance never cancels.
- Bound check: the speed is at least the largest |component| and at most their sum.

**Example:** A particle has $x(t) = t^2$, $y(t) = t^3$. Find its speed at $t = 1$.
**Step 1.** Speed is the magnitude of the velocity vector, so both derivatives get squared before anything is added — adding them first is the classic slip.
**Step 2.** $x' = 2t$ and $y' = 3t^2$, so at $t = 1$ the speed is $\sqrt{(2 \cdot 1)^2 + (3 \cdot 1^2)^2} = \sqrt{13}$.
The speed is **3.61**.

**Try it:** For the same path $x(t) = t^2$, $y(t) = t^3$, find the speed at $t = 2$, to the nearest hundredth.
check: **12.65**

## Skill 3 — Choose polar limits from where $r$ vanishes

Area swept by the radius: $A = \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta} r^2\,d\theta$, where $[\alpha, \beta]$ traces the region **exactly once**.

- Solve $r = 0$ first: two consecutive solutions bound one petal or loop.
- $r = a\cos(k\theta)$ has $k$ petals when $k$ is odd and $2k$ when $k$ is even — and a full turn retraces an odd rose twice.
- Useful integral: $\cos^2 u = \dfrac{1+\cos 2u}{2}$.

**Example:** Find the area of one petal of $r = 2\cos(2\theta)$.
**Step 1.** A petal runs between consecutive zeros of $r$, so solve $r = 0$ before setting any limits: $\cos 2\theta = 0$ gives $\theta = \pm\frac{\pi}{4}$.
**Step 2.** $A = \frac{1}{2}\displaystyle\int_{-\pi/4}^{\pi/4}(2\cos 2\theta)^2 d\theta = 2\int_{-\pi/4}^{\pi/4}\cos^2 2\theta\,d\theta$.
One petal has area $\boldsymbol{\frac{\pi}{2}}$.

**Try it:** Find the area of one petal of $r = 3\cos(2\theta)$, in exact form.

check: $\boldsymbol{\frac{9\pi}{8}}$

**(!) Watch out:** A negative speed, a slope evaluated at a coordinate, and a polar area taken over 0 to $2\pi$ by default are the three errors that produce clean-looking numbers. Each is caught by one question: is this a magnitude, is this a $t$, does this sweep the region once?